

Stochastic & Deterministic Q-Learning

REINFORCEMENT LEARNING TECHNICAL NOTES

1. Core Concept: Deterministic vs. Non-Deterministic Environments

In Reinforcement Learning, the nature of environmental state transitions and reward structures significantly impacts the convergence behavior of control strategies:

- **Deterministic Environments:** Taking a specific action a in a given state s will always result in a transition to the exact same next state s' and yield an identical immediate reward R .
- **Non-Deterministic (Stochastic) Environments:** Taking an action a in state s can yield probabilistic outcomes. It may lead to a set of potential next states s' or generate variable rewards according to a joint transition probability distribution function, expressed as $P(s', R | s, a)$.

Because individual transitions in a stochastic environment contain inherent variance, an agent cannot rely completely on a single sample step to represent true value. To handle this, Q-learning incorporates a temporal difference learning rate parameter (α) to smoothly maintain a rolling expected average over historical iterations.

2. The Q-Learning Update Equation

The temporal difference update rule allows model-free agents to iteratively refine value estimates based on real-time experiential feedback:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Mathematical Components Definition:

- $Q(s, a)$: The current state-action quality value representing the expected cumulative discounted future reward when initiating action a from state s .
- $\leftarrow$: The assignment operator updating the historical matrix metric with the newly computed value estimate.
- α (Learning Rate): A tuning factor controlling the scaling coefficient of new observations relative to existing data thresholds ($0 < \alpha \leq 1$). In highly non-deterministic systems, a lower value (e.g., $\alpha = 0.1$) is critical to suppress ambient noise.
- R : The immediate scalar reward feedback returned by the environment transition wrapper.
- γ (Discount Factor): A fractional value determining the present value of future long-term payoffs ($0 \leq \gamma < 1$). Values close to 0 prioritize short-term returns; values approaching 1 weigh distant future payoffs.

- $\max_a, Q(s', a')$: The optimal estimated potential value obtainable in the successor state s' by choosing the action maximizing the target Q-distribution profile.
- $[R + \gamma \max_a, Q(s', a') - Q(s, a)]$: The **Temporal Difference (TD) Error**. This tracks the explicit mathematical delta between the newly observed destination target value profile and the baseline reference.

3. Step-by-Step Solved Problem Scenario

Consider a simplified discrete finite Grid World system consisting of a small environment state map to trace a manual update loop calculation.

Problem Parameters:

- **Current State (s)**: State A
- **Selected Action (a)**: "Move Right"
- **Configured Hyperparameters**: Learning Rate $\alpha = 0.5$, Discount Factor $\gamma = 0.8$

Initial Q-Table Values Matrix:

State Matrix Index	Action: "Move Left"	Action: "Move Right"
State A (s)	0.0	2.0 (Active Q Value)
State B (s')	4.0	5.0

Stochastic Transition Event Trigger:

The agent initiates the "Move Right" command from State A. Due to environmental stochasticity (e.g., surface slickness or noise friction), the agent experiences an unexpected slide transition, instead updating spatial coordinates to settle in **State B (s')** while receiving an immediate objective scalar reward value of $R = 3$.

Algorithmic Resolution Sequence:

Step 1: Parse and Map Known Environmental Variables

Baseline $Q(s, a) = Q(\text{ext}\{\text{State A}\}, \text{ext}\{\text{"Move Right"}\}) = 2.0$

Reward $R = 3$, Learning Rate $\alpha = 0.5$, Discount Factor $\gamma = 0.8$

Step 2: Determine Optimal Successor Future Estimate Value

Evaluate maximum yield choices over State B parameters: $\max(4.0, 5.0) = 5.0$

Hence, $\max_a, Q(s', a') = 5.0$

Step 3: Calculate the Temporal Difference Target Value

$$\text{ext}\{Target\} = R + \gamma \max_{a'} Q(s', a')$$

$$\text{ext}\{Target\} = 3 + (0.8 \text{ times } 5.0) = 3 + 4.0 = 7.0$$

Step 4: Compute the Resulting TD Error Deviation Value

$$\text{ext}\{TD \text{ Error}\} = \text{ext}\{Target\} - \text{ext}\{Old\} \text{ } Q(s,a)$$

$$\text{ext}\{TD \text{ Error}\} = 7.0 - 2.0 = 5.0$$

Step 5: Apply Learning Rate Scale Adjustment to Update Q-Table

$$Q(s, a) \leftarrow \text{ext}\{Old\} \text{ } Q(s,a) + \alpha \text{ times } (\text{ext}\{TD \text{ Error}\})$$

$$Q(\text{ext}\{State A\}, \text{ext}\{ "Move Right" \}) \leftarrow 2.0 + 0.5 \text{ times } (5.0) = 2.0 + 2.5 = 4.5$$

Final Computational Conclusion:

Following execution of the policy evaluation sequence, the localized historical index mapping value for taking the action "Move Right" while localized within State A securely increments from its base value of **2.0** to a corrected output value of **4.5**.